

Case Study: Modernizing the National Medical Association's Technology & Workspace

Client Overview

The **National Medical Association (NMA)** is the nation's oldest and largest organization representing African American physicians and their patients. Committed to advancing health equity, NMA recognized the need to bring its operations and technology in line with its modern mission and forward-looking leadership vision.

Beginning in **2016**, NMA partnered with **DET resources** to transform its aging IT systems into a secure, cloud-first, and culturally inspired digital environment that reflects both innovation and inclusivity.



Challenge

By 2016, the NMA's IT systems were running on **outdated, end-of-life infrastructure** – including servers, switches, and storage platforms that were well beyond their useful lifespan. These legacy systems limited performance, posed security risks, and constrained remote access for staff and members.

As the organization grew, so did the need for:

- **A reliable and secure infrastructure** to replace aging hardware.
- **Modern collaboration tools** for hybrid work.
- **A unified digital environment** that aligns with the organization's culture and values.
- **An office technology experience** that blended function with aesthetic design – embracing art, culture, and inclusivity.

DET's long-term partnership evolved through three key modernization phases.



Phase 1: Foundational Server Modernization (2016)

The relationship between DET and the NMA began with an urgent need to **stabilize and modernize legacy systems** that had been in operation for more than a decade.

Scope of Work

- **Replaced legacy physical servers** with a modern, virtualized server environment to improve reliability and data performance.
- **Upgraded storage and security systems**, reducing downtime and data loss risks.
- **Implemented structured backup and redundancy processes** for business continuity.

Outcomes

- Improved system performance and operational reliability.
- Extended lifespan of critical applications and data repositories.
- Established a foundation for future cloud migration efforts.



Phase 2: Cloud Infrastructure Transformation (2023)

Building on the earlier modernization, DET led the **transition to a fully cloud-based infrastructure** to eliminate dependency on on-premises hardware and enhance security, mobility, and scalability.

01	02	03
Server Migration Decommissioned legacy MARS servers, migrating all user data to Microsoft OneDrive and SharePoint Online.	Data Archival Transitioned accounting data to secure cold cloud storage for archival access.	Network Infrastructure Deployed Cisco Meraki firewalls, switches, and wireless access points to support a secure hybrid network environment.

04	05
Security Assessment Performed comprehensive cybersecurity assessments, including penetration testing, vulnerability scans, and IOC (Indicator of Compromise) checks.	System Configuration Configured Microsoft 365 backup and SMTP printing solutions, enabling resilience and simplified operations.

100% cloud migration Removing the need for any physical servers.	Enhanced security posture Verified through penetration testing and vulnerability mitigation.
Unified digital workplace Improving collaboration and data accessibility for staff nationwide.	Cost optimization Reducing IT overhead and maintenance costs by over 40%.

Phase 3: Cultural and Workspace Modernization (2024)

As part of NMA's move to a new headquarters, DET was re-engaged to **design and implement a modern low-voltage and audiovisual infrastructure** that mirrored the organization's evolving identity – one that merges medicine, technology, art, and culture.

Scope of Work

- **Developed complete low-voltage and data cabling plans** – including installation, labeling, certification, and testing of all Cat6 data drops and IT room cabling.
- **Designed and implemented a new sound masking system** to enhance speech privacy and reduce distractions in open areas.
- **Integrated high-end audiovisual technology** using **Samsung Frame displays** throughout the workspace. These displays double as digital art canvases, blending seamlessly into the office aesthetic to reflect the NMA's commitment to art, culture, and inclusivity.
- **Installed advanced conferencing and collaboration tools**, including Shure ceiling microphones, Yealink 4K intelligent cameras, and Barco ClickShare wireless conferencing systems.
- **Provided full system programming and integration**, ensuring every room supported efficient hybrid collaboration.

Outcomes

- **A modern, inclusive, and inspiring workspace** that celebrates both technology and culture.
- **Enhanced meeting and collaboration experiences** through integrated, easy-to-use audiovisual systems.
- **Optimized cabling and IT infrastructure**, supporting scalability and future upgrades.
- **Art-infused technology design**, aligning the physical environment with NMA's vision of excellence and cultural pride.

Final Architecture

Layer	Technology / Solution
Network Core	Cisco Meraki Firewall, Switches, and Access Points
Cloud Platform	Microsoft 365 (SharePoint, OneDrive, Teams, Exchange)
Security	Vulnerability and IOC Testing, Multi-Factor Authentication
AV & Collaboration	Samsung Frame Displays, Shure Audio, Yealink Cameras, Barco ClickShare
Sound & Privacy	Multi-zone Sound Masking System
Data Infrastructure	Certified Cat6 Cabling and Cloud-managed Network Racks

Key Results

Objective	Result
Stabilize and modernize aging servers	✔ Completed in 2016
Transition from on-premises to cloud	✔ Achieved in 2023
Integrate culture-driven office technology	✔ Completed in 2024
Enhance cybersecurity and compliance	✔ Verified via penetration testing
Create inclusive, art-inspired workspace	✔ Fully realized in new NMA headquarters



Conclusion

The National Medical Association's multi-year transformation with **Diverse Excellence Technology (DET)** demonstrates how thoughtful modernization can bridge **technology, culture, and community**.

From **server upgrades in 2016** to **full cloud adoption in 2023** and a **visually inspiring AV-integrated headquarters in 2024**, DET has delivered scalable, secure, and aesthetically aligned solutions that embody the NMA's mission of excellence, innovation, and inclusivity.